

The A/B-theory debate, temporal ontology, and spacetime

PHIL UA-78 Metaphysics

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The biggest divide in the philosophy of time (and arguably in metaphysics) is between the A-theory of time and the B-theory of time.

The debate has been around for a long time and gets explained in different, more or less metaphorical ways; but I propose that the key disagreement is that B-theorists accept, while A-theorists deny:

Fact Eternalism

Every state of affairs that ever obtains, always obtains.

Properties and change

Properties

Throughout this course we have been using 'property' and 'state of affairs' in tightly connected ways. For any object and property, there is the state of affairs we get by plugging the object into the property. For the object to instantiate the property is just for this state of affairs to obtain.

So, there is no way it could obtain eternally without the object having the property eternally. That means the B-theory would seem to entail:

Property Eternalism

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For every object and property, if the object ever instantiates the property, it always instantiates the property.

Worry for B-theorists: isn't this obviously false? Isn't it manifestly true that objects often gain and lose properties? For example, when I sit down I gain the property of *sitting*.

- For example, Miller (p. 26), although a B-theorist, treats it as uncontroversial that for something to change it must have different properties at different times, so that Property Eternalism would imply the obviously false conclusion that nothing changes.

How should B-theorists respond? As always, since their guiding idea is 'Time is like space', they should consider the spatial analogue of Property Eternalism:

Property Ubiquitism

For every object and property, if the object instantiates the property somewhere, it instantiates the property everywhere.

Again one might worry that this is obviously false. Isn't it obvious that there are things that have properties in some places that they lack in other places? For example, *being popular* is a property Trump has in Kentucky but not in Massachusetts.

A proposed response on behalf of the B-theorist

'Property' in English doesn't always mean what we've been using it to mean in this metaphysics course.

Sometimes, 'property' is instead applied to (some of) the things that we have been calling *relations*.

- ▶ 'The property of being popular' can stand for the binary popular-in relation (between people and places)
- ▶ 'The property of sitting' can stand for the binary sitting-at relation (between people and times).

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I’m not suggesting that these are the only possibilities—you *could* also use ‘the property of sitting’ to refer to the property of sitting at t (where t is the time you’re talking at), or ‘the property of being popular’ to refer to the property of being popular in NYC, or popular among the people who know you, or. . . .

Change in the A-theory

Here's a tempting theory of change for A-theorists (using 'property' in our ordinary way):

Simple Theory of Change

Object x changes from t_1 to t_2 iff there is some property that x has at t_1 and lacks at t_2 (or vice-versa).

But note that this makes the claim that x changes from t_1 to t_2 almost trivial! If one thing changes, everything does: supposing x is F at t_1 and not F at t_2 : then every other object y is *such that* x is F at t_1 and *not such that* x is F at t_2 .

And even if everything 'freezes' starting at t_1 , they will still be changing as regards properties of the form *having been frozen for so many seconds*.

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Moral: A-theorists should say that normally, when we say that something changes, we are restricting the quantifier in the Simple Theory of Change to some *restricted range*

Change in the B-theory

Here is Bertrand Russell, an arch B-theorist:

Change is the difference, in respect of truth or falsehood, between a proposition concerning an entity and the time T , and a proposition concerning the same entity and the time T' , provided that these propositions differ only by the fact that T occurs in the one where T' occurs in the other. (Russell, *The Principles of Mathematics*, §442)

This theory—called the ‘Cambridge theory of change’—is subject to the same kind of objections as the Simple Theory of Change in an A-theoretic setting. For any object x and any distinct times t_1 and t_2 , there is always at least one relation it bears to t_2 but not to t_1 , namely *being an object and a time such that the first is x and the second is t_2* .

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Moral: the B-theorist should say that normally, when we say that something changes from one time to another, we are restricting our attention to some restricted range of

Temporal ontology

A choice point for A-theorists: Permanentism

Permanentism

Always, everything is such that always, it is identical to something.

where the quantifiers are *unrestricted*.

- Assuming the Logical Analysis of Existence, this can be restated as 'Always, everything always exists'

Temporaryism is the negation of Permanentism.

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I take the B-theory to imply Permanentism. 'Always' is redundant unless there is a hidden variable for it to bind. And in 'it is identical to something', with 'something' understood as unrestricted, there is no such hidden variable.

By contrast, A-theorists can go either way on Permanentism.

There's a classic theory—the *Moving Spotlight Theory*—that endorses the A-theory together with Permanentism. (But maybe there's more to it than that?)

Varieties of temporaryism

Temporaryists must reject one or both of the following claims, which together entail Permanentism (since 'it has forever been that P, and P, and it will forevermore be that P' entails 'always P'):

Nothing Lost

Always, everything is such that it will forevermore be identical to something.

Nothing Gained

Always, everything is such that it has forever been identical to something.

There's a classic theory—the *Growing Block Theory*—that endorses the A-theory and Nothing Lost, but rejects Nothing Gained (and hence Permanentism). (But maybe there's more to it than that?)

There's also a classic theory—*Presentism*—that endorses the A-theory and rejects both Nothing Lost and Nothing Gained. (But maybe there's more to it than that?)

Let's continue to use 'concrete' in Williamson's way.

- Think of it as meaning 'has a shape' (or 'has a size' or 'has a mass'...)—though if you're a theist or Cartesian you will need to be a bit more careful.

Philosophers of time like to apply the words 'past', 'present', and 'future' to objects like people, not just to times and events.

This has always sounded weird to me! But I suggest that **when we are talking about objects** (as opposed to *times* and *events*), we can understand this usage as follows:

x is present $:=$ x is concrete

x is past $:=$ x has been concrete, but x is not concrete

x is future $:=$ x will be concrete, but x is not concrete

What is Presentism?

Presentists are supposed to think that *everything is present*, hence *nothing is past* and *nothing is future*.

Proposal: presentism is

Temporal Concreteness Essentialism

Always, every concrete object is always such that if it's identical to something, it's concrete.

Uncontroversially, there are concrete objects that have not always been concrete and will not always be concrete. So Presentism entails that Nothing Lost and Nothing Gained are both false.

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- Note that 'present' does not mean 'concrete' when applied to *times*. So presentism as I am understanding it does not imply that there is no time other than the present time. (Which is good, since that seems absurd.)
- It's not so obvious what we should say about *events*. Are they more like objects?

What is the Growing Block View?

Growing Blockers are supposed to think that *everything is either past or present*.

Proposal: the Growing Block is the conjunction of Nothing Lost with

Temporal One-sided Concreteness Essentialism

Always, every concrete object is always such that if it's identical to something, either it is concrete or it has been concrete.

- ▶ Again, this doesn't have to deny that there are future *times*, since 'future' for times is not defined in terms of 'concrete'.
- ▶ Given how we defined 'future', this does not entail 'No object is future', but it does entail 'Every future object is also past'.